

MOLECULAR AND STRUCTURAL INSIGHTS INTO *HAEMOPHILUS INFLUENZAE* HAEM ACQUISITION FROM HUMAN HAEMOPROTEINS

Xiu TIAN¹, Pierre LEGRAND², Heddy SOUFARI², Eric LARQUET², Patrick ENGLAND³, Sylvianne HOOS³, Alexandre POZZA¹, Cain TRAN¹, Laurent CATOIRE¹, Philippe DELEPELAIRE¹, Valérie BIOU¹, Nathalie DAUTIN¹

¹ Institute of Physico-Chemical Biology (IBPC), UMR7099, Laboratory of Membrane Proteins Biochemistry, CNRS, Paris Cité University, Paris, France

² Synchrotron Soleil, POLARIS facility, L'Orme les Merisiers, Saint-Aubin, France

³ Molecular Biophysics Platform, C2RT, Pasteur Institute, Université Paris Cité, CNRS, UMR 3528, Paris, France.

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ABSTRACT

Haemophilus influenzae is an opportunistic, Gram-negative bacterium that only colonizes humans and can be responsible for various diseases. As a haem auxotroph, this bacterium needs to acquire haem from its host haemoproteins in order to survive and sustain infection. Hgps (haemoglobin and haemoglobin-haptoglobin proteins) are a family of TonB-dependent transporters that bind haemoglobin (Hb) and are essential for *H. influenzae* growth in the presence of haemoglobin-haptoglobin (Hb-Hp) as sole source of haem. Here, we used a combination of functional analysis and cryo-EM to gain insight into the mechanism by which Hgps bind haemoglobin (Hb) and extract haem from it. We found that Hgps preferentially interact with human, dimeric haemoglobin ($K_D \sim \text{nM}$). This interaction results in the distortion of the F-helix from the haemoglobin beta chain (Hb- β) and leads to the energy-independent transfer of haem from Hb- β to a conserved tyrosine located on the receptor plug-loop. Following transfer, the Tyr-Fe bond is stabilized by the Hb- β proximal histidine (His92, F8), that has been displaced 8.5Å toward the receptor. Altogether, we described a new mechanism of haem acquisition in Gram-negative bacteria, which deviates significantly from the one described previously for haem-TBDTs and appears more similar to the one described for Gram positive bacteria haem import systems.